

Nithilan S

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EDUCATION

Sri Sivasubramaniya Nadar College of Engineering, Chennai

B.Tech. Information Technology — Rank: 13/141

Aug 2024 – Present

CGPA: **8.511**

Maharishi Vidya Mandir Senior Secondary School, Chennai

CBSE Class XII & X

Jun 2020 – May 2024

Percentage: **94.8%**

TECHNICAL SKILLS

Languages: Python, Java, C, Javascript, SQL

Systems & Backend: Distributed systems, concurrency, multithreading, synchronization, REST APIs, FastAPI, arq, Redis

Infra & Reliability: Docker, Docker Compose, caching, rate limiting, load testing (Locust), pytest, Git

AI / ML: RAG, LLM evaluation, semantic retrieval, FAISS, BM25, sentence-transformers

Databases: MySQL, Oracle SQL, SQLite

EXPERIENCE

Software Engineering Intern

May 2026 – Jun 2026

QPay India Pvt Ltd

Chennai, India

- Built a **payment-gateway risk-scoring engine** (Python, FastAPI, React) producing an explainable 0–100 anomaly score from per-account behavioural baselines, framed as a deviation score (no fraud labels) with low-confidence flagging for sparse histories.
- Built a review **sentiment classifier** benchmarking TF-IDF, GloVe, and fine-tuned BERT on 1.8M reviews (best **F1 0.82**); found raw text beat preprocessed for BERT, served via FastAPI.

Research Intern

May 2026 – Present

National Institute of Technology, Tiruchirappalli (NIT Trichy)

Tiruchirappalli, India

- Researching **LLM reasoning** and **retrieval-augmented QA** under Dr. A. Santhanavijayan; scoping **step-level speculative reasoning** for multi-hop QA with gold-grounded step evaluation (NLI over HotpotQA/MuSiQue).

Software Testing Intern

Jun 2025 – Jul 2025

CUMI (Carborundum Universal Limited)

Chennai, India

- Automated **UI/frontend regression testing** (SikuliX + Python), replacing manual click-through verification.

PROJECTS

High-Throughput Asynchronous Job-Processing Service

FastAPI, arq, Redis, Docker Compose, Locust, Prometheus

- Designed an async, horizontally-scalable distributed job-processing system (FastAPI + arq + Redis worker pool) sustaining **446 req/s** with **p99 latency 120 ms** and zero failures under 50 concurrent users.
- Engineered concurrency-safe reliability machinery — idempotency (atomic SET NX), token-bucket rate limiting (atomic Lua), caching, dead-letter queue with exponential-backoff retries — each guarantee verified by a **13-test** suite; near-linear scaling to **1033 jobs/s** (3 workers).

Natural-Language-to-SQL System with Evaluation Harness

Python, FastAPI, sentence-transformers, Llama-3.3-70B (Groq), SQLite, React

- Built a retrieval + LLM pipeline (FastAPI, React) translating natural language to runnable SQL with a read-only AST safety guard, a bounded self-correction loop, and structured error handling; **85% execution accuracy** on a 40-question eval set.
- Engineered a resumable, checkpointed **evaluation harness** with a controlled ablation to regression-test pipeline accuracy across components and providers.

Filing Intelligence — Document QA Service

FastAPI, React, FAISS, BM25, Redis, arq, Gemini, Docker

- Built a full-stack document-QA service (FastAPI + React) with **async ingestion** (arq) and a two-stage Redis cache serving warm requests without invoking retrieval or the LLM, plus inline source-cited responses.
- Designed a hybrid retrieval service (FAISS + BM25 + Rank Fusion) over 1,911 chunks behind a REST API at **3.7 ms mean**; eval harness (recall@k, MRR) justified the shipped mode (**MRR 0.56** vs 0.41 dense).

LEADERSHIP & ACTIVITIES

Core Member, Open Source Software Club — SSN Coding Club

SSN College of Engineering

- Co-organized FunOverflow, a college-wide non-technical event featuring three rounds (Sept 2025).